「Continuous.

Let $f: X \to Y$ be a map where X, Y are Metric spaces, and $a \in X$. TFAE:

a) f is continuous at a

b) If a sequence $\{x_n\} \subseteq X$ converges to a, then $\{f(x_n)\}$ converges to $f(a)$.

Proof. Suppose f is continuous at a. Let $\varepsilon > 0$ be given. Then, there exists $\delta > 0$ s.t

$$x \in B_\delta(a) \Rightarrow f(x) \in B_\varepsilon(f(a))$$

by definition. But $N \in \mathbb{N}$ s.t

$$n \geq N \Rightarrow x_n \in B_\delta(a)$$

Now, for $n \geq N$,

$$f(x_n) \in B_\varepsilon(f(a))$$

To show the converse, using contraposition. Suppose that f is not continuous at a. That is, there exists $\varepsilon > 0$ s.t for any $\delta > 0$, there is $x \in B_\delta(a)$ satisfying

$$d(f(x), f(a)) \geq \varepsilon.$$

Therefore, for each $n \in \mathbb{N}$, we can choose $x_n \in B_{\frac{1}{n}}(a)$ and $d(f(x_n), f(a)) \geq \varepsilon$.

By construction, $\{x_n\}$ converges to a but $f(x_n)$ does not converges to $f(a)$.

「Thomae's function.

Define a map

$$T: (0, \infty) \to \mathbb{R}: x \mapsto \begin{cases} \frac{1}{p} & x = \frac{q}{p}, \text{ irreducible form (or } \gcd(p,q) = 1) \\ 0 & x \notin \mathbb{Q} \end{cases}$$

Then T is continuous at every irrational number, but not continuous at rationals.

Proof. Let $c \in (0, \infty)$ be an irrational number. Given $\varepsilon > 0$, there is $N \in \mathbb{N}$ s.t $\frac{1}{N} < \varepsilon$.

Since there exist only finite number rationals such that the denominator smaller than N, more specifically, $S = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \dots, \frac{N-1}{N}\right\}$ (at most $\frac{N(N-1)}{2}$)

we can choose small enough $\delta > 0$ s.t $(c-\delta, c+\delta) \cap S = \emptyset$. Now,

$$|x - c| < \delta \Rightarrow |f(x) - f(c)| < \frac{1}{n} < \varepsilon$$

Therefore, f is continuous at irrationals. But, let $x \in (0, \infty)$ be a rational.

Denote $x = \frac{q}{p}$ where $p, q \in \mathbb{N}$, $\gcd(p,q) = 1$. Taking $\varepsilon = \frac{1}{p}$. Then the density of irrationals gives

$$\forall \delta > 0, \quad \exists \text{ irrational } t \in (x - \delta, x + \delta) \text{ s.t } |f(x) - f(t)| = \frac{1}{p}$$

Thus, not continuous at irrationals.

「Exercise.

20.E Define a map

$$f:\mathbb{R}\to\mathbb{R} : x\mapsto \begin{cases} x & \text{if } x\notin\mathbb{Q} \\ 1-x & \text{if } x\in\mathbb{Q} \end{cases}$$

Then f is continuous only at $x=\frac{1}{2}$.

Sol). Let $\varepsilon>0$ be given. Taking $\delta=\varepsilon$. Then for any $x\in(\frac{1}{2}-\delta,\frac{1}{2}+\delta)$,

$$\left|f(x)-\frac{1}{2}\right| = \begin{cases} \left|x-\frac{1}{2}\right| & \text{if } x\notin\mathbb{Q} \\ \left|1-x-\frac{1}{2}\right| & \text{if } x\in\mathbb{Q} \end{cases}$$

Both case satisfies $\left|f(x)-\frac{1}{2}\right|<\varepsilon$. Therefore continuous.

Let $x\in\mathbb{R}$ with $x\neq\frac{1}{2}$. If $x\in\mathbb{Q}$, then by density of rationals, we can construct $\{x_n\}$ in $\mathbb{R}\setminus\mathbb{Q}$ s.t. $x_n\to x$ as $n\to\infty$

But $f(x_n)=x_n\to x$ as $n\to\infty$, not converges to $f(x)=1-x$.

Similarly, if $x\notin\mathbb{Q}$, it can be shown by the same process.

20.F If a continuous map $f:\mathbb{R}\to\mathbb{R}$ satisfies

$$f(x)=0 \quad \text{for all } x\in\mathbb{Q}$$

Then f is zero map.

Proof. Suppose that f is not zero map. Given $x\in\mathbb{R}$ satisfying $f(x)\neq 0$, take $\varepsilon=|f(x)|$.

For any $\delta>0$, by the density of rationals,

$$\exists t\in(x-\delta,x+\delta) \text{ s.t. } |f(x)-f(t)|=|f(x)|>\frac{1}{2}|f(x)|.$$

It is contradiction.

20.G Let $f,g:\mathbb{R}\to\mathbb{R}$ be continuous map. If $f(x)=g(x)$ $\forall x\in\mathbb{Q}$, then $f=g$.

Proof. Take $h=f-g$ and apply 20.F.

20.H. Using $|\sin x| \le |x|$, show sine function is continuous at $x=0$.
Moreover, using

$$\sin x - \sin u = 2\sin\tfrac{1}{2}(x-u)\cos\tfrac{1}{2}(x+u)$$

Show sine function continuous on $\mathbb{R}$.

proof. Given $\varepsilon>0$, take $\delta=\varepsilon$. Then,

$$|x|<\delta \Rightarrow |\sin x| \le |x| < \delta = \varepsilon$$

Thus, continuous at $x=0$. Similarly, taking $\delta=\varepsilon$ then

$$|x-u|<\delta \Rightarrow |\sin x - \sin u| = \left|2\sin\tfrac{1}{2}(x-u)\cos\tfrac{1}{2}(x+u)\right| \le |x-u| < \varepsilon.$$

20.I. Define a map

$$g:\mathbb{R}\to\mathbb{R}: x \mapsto \begin{cases} x\cdot\sin\left(\frac{1}{x}\right) & \text{if } x\neq 0 \\ 0 & \text{if } x=0 \end{cases}$$

Then g is continuous on $\mathbb{R}$.

proof. Since x, $\sin x$, $\frac{1}{x}$ are continuous if $x\neq 0$, so is $x\sin\frac{1}{x}$.
Given $\varepsilon>0$, put $\delta=\varepsilon$. Then

$$0<|x|<\delta \Rightarrow \left|x\sin\left(\tfrac{1}{x}\right)\right| \le |x| < \varepsilon$$

Therefore, g is continuous everywhere.

20.K. Define a map

$$F:\mathbb{R}^2\to\mathbb{R}: (x,y)\mapsto \begin{cases} x^2+y^2 & \text{if } (x,y)\in\mathbb{Q}\times\mathbb{Q} \\ 0 & \text{otherwise} \end{cases}$$

Determine the points where F is continuous.

Sol) Consider $(0,0)$. Given $\varepsilon>0$, taking $\delta=\sqrt{\varepsilon}$

$$(x,y)\in B_\delta(0,0) \Rightarrow |f(x,y)| \le x^2+y^2 < \delta^2 = \varepsilon.$$

Thus conti. at zero. But, if $(x,y)\neq 0$, regardless of whether $(x,y)\in\mathbb{Q}\times\mathbb{Q}$ or not, we can construct a sequence $(x_n,y_n)\to(x,y)$ but does not $f(x_n,y_n)\to f(x,y)$.

20.L. A map $f: \mathbb{R} \to \mathbb{R}$ is called additive if:

$$\forall x, y \in \mathbb{R}, \quad f(x+y) = f(x) + f(y). \qquad f(2) = f(1) + f(1) = 2f(1) \qquad f\left(\frac{1}{m}\right) = \frac{1}{m} f(1)$$

If additive map is continuous at $x=0$, then it is continuous at every point.

Moreover, monotone additive map is continuous everywhere.

Sol). Given $\varepsilon > 0$, there is $\delta > 0$ s.t

$$|t| < \delta \Rightarrow |f(t)| < \varepsilon \qquad (f(0) = 0, \text{ because } f(0) = 2f(0))$$

since f is continuous at $x=0$. Now, given $x \in \mathbb{R}$,

$$|x-t| < \delta \Rightarrow |f(x) - f(t)| = |f(x-t)| < \varepsilon \qquad (\text{since } f(x-t) + f(t) = f(x))$$

by additivity.

Meanwhile, suppose that f increases monotonically, and additive.

Given $\varepsilon > 0$, take $c \neq 0$ s.t $f(c) \neq 0$ (If no such c exists, then $f \equiv 0$, there is nothing to prove)

observe that $f(t) = f(\frac{1}{2}t) + f(\frac{1}{2}t) = 2 \cdot f(\frac{1}{2}t)$ for any $t \in \mathbb{R}$. Inductively, for all $n \in \mathbb{N}$,

$$\frac{1}{2^n} f(t) = f\left(\frac{1}{2^n} t\right)$$

Take $N \in \mathbb{N}$ satisfying $\frac{1}{2^N} f(|c|) < \varepsilon$. That is, $\frac{1}{2^N} f(|c|) = f\left(\frac{1}{2^N}|c|\right) < \varepsilon$

Now, $|x| < \frac{1}{2^N}|c| \Rightarrow |f(x)| = f(|x|) < f\left(\frac{1}{2^N}|c|\right) < \varepsilon$

$0 = f(x-x) = f(x) + f(-x)$ $\qquad$ f increases

Therefore f is continuous at $x=0$, so is continuous everywhere.

「20.M. Additive continuous map $f: \mathbb{R} \to \mathbb{R}$ must be $x \mapsto f(1)x$.

Define a map $\lceil \cdot \rceil : \mathbb{R} \to \mathbb{Z} : x \mapsto \min\{n \in \mathbb{Z} \mid n \geq x\}$ (It is well-defined by well-ordering principle)

And observe that for any $x \in \mathbb{R}$,

$$\lim_{n \to \infty} \frac{\lceil 2^n x \rceil}{2^n} = x \qquad \text{because} \qquad 2^n x \leq \lceil 2^n x \rceil < 2^n x + 1.$$

Given $x \in \mathbb{R}$, $f\left(\frac{\lceil 2^n x \rceil}{2^n}\right) = \frac{\lceil 2^n x \rceil}{2^n} \cdot f(1)$ by additivity. Now,

$$cx = f(1) \cdot x = f(1) \cdot \lim_{n \to \infty} \frac{\lceil 2^n x \rceil}{2^n} = \lim_{n \to \infty} f\left(\frac{\lceil 2^n x \rceil}{2^n}\right) = f\left(\lim_{n \to \infty} \frac{\lceil 2^n x \rceil}{2^n}\right) = f(x)$$

I love dyadic rational numbers. $\qquad$ f. Conti.

20.N If $g:\mathbb{R}\to\mathbb{R}$ satisfies that

$$g(x+y) = g(x)\cdot g(y) \quad \text{for all } x,y\in\mathbb{R}$$

then g has the following properties;

1). If g is continuous at $x=0$, then it is continuous everywhere

2). If $g(x)=0$ for some $x\in\mathbb{R}$, then $g=0$.

Proof If $g(x)=0$ for some $x\in\mathbb{R}$, then for any $y\in\mathbb{R}$,

$$g(y) = g(y-x+x) = g(y-x)\,g(x) = 0.$$

The zero map is clearly continuous, suppose that $g(x)\neq 0$ $\forall x\in\mathbb{R}$.

Let $a\in\mathbb{R}$ be given. Given $\varepsilon>0$, there exists $\delta>0$ s.t

$$|t|<\delta \Rightarrow |g(t)-g(0)| = |g(t)-1| < \frac{\varepsilon}{|g(a)|} \quad \left(g(0+0)=g(0)^2 \text{ implies } g(0)=1\right)$$

Now, $|x-a|<\delta \Rightarrow |g(x)-g(a)| = |g(a)|\cdot|g(x-a)-1| < \varepsilon.$

$g(x)=g(x-a+a)=g(x-a)\cdot g(a)$

Doodles.

Observe that $g(0+0)=g(0)\cdot g(0)$ implies $g(0)=0$ or 1.

If $g(0)=0$, then for any $x\in\mathbb{R}$, $g(x)=g(x+0)=g(x)g(0)=0$.

Clearly zero map is continuous, suppose that $g(0)=1$.

Given $\varepsilon>0$, there exists $\delta>0$ s.t

$$|t|<\delta \Rightarrow |g(t)-g(0)| = |g(t)-g(1)| < \varepsilon$$

Since the assumption of 1).

Given $\varepsilon>0$, there is $\delta>0$ s.t

$$|t|<\delta \Rightarrow |g(t)-g(0)|<\varepsilon.$$

Now, let $a\in\mathbb{R}$ be given. Then

$g(x)=g(x-a+a)=g(x-a)\cdot g(a)$

$$|x-a|<\delta \Rightarrow |g(x)-g(a)| = |g(a)|\cdot|g(x-a)-1| \le |g(x-a)-g(0)| < \varepsilon$$

$g(0+0)=g(0)^2 \Rightarrow g(0)=0$ or 1

if $g(0)=0$, then $g(a)=g(a)\cdot g(0)=0$

if $g(0)=1$,

20.O False. Ask a toddler on the street.

But the converse is true, by the triangle inequality.

20.P. Let $f, g : \mathbb{R}^p \to \mathbb{R}$ be continuous at $a \in \mathbb{R}^p$, and define

$$h : \mathbb{R}^p \to \mathbb{R} : x \mapsto \sup\{f(x), g(x)\} \quad \text{and} \quad K : \mathbb{R}^p \to \mathbb{R} : x \mapsto \inf\{f(x), g(x)\}$$

Prove h, K are continuous at $x = a$.

Proof. Since $\sup\{f(x), g(x)\} = \frac{1}{2} \cdot \left(f(x) + g(x) + |f(x) - g(x)|\right)$

directly h is continuous at a, being compositions of continuous.

20.Q Ask...

20.R. If $f : I \to \mathbb{R}$ is monotone increasing, then TFAE:

a). f is continuous at $c \in I^\circ$

b) $\forall \varepsilon > 0$, $\exists x_1, x_2 \in I$ with $x_1 < c < x_2$ s.t. $f(x_2) - f(x_1) < \varepsilon$.

c) $\sup\limits_{x < c} f(x) = f(c) = \inf\limits_{x > c} f(x)$

Proof. Suppose f is continuous at $c \in I^\circ$. Given $\varepsilon > 0$, there is $\delta > 0$ s.t.

$$|x - c| < \delta \Rightarrow |f(x) - f(c)| < \frac{\varepsilon}{2}$$

Take $x_1 = c - \frac{\delta}{2}$, $x_2 = c + \frac{\delta}{2}$. Then clearly $|x_i - c| = \frac{\delta}{2} < \delta$ $(i = 1, 2)$, and

$$f(x_2) - f(x_1) = |f(x_2) - f(x_1)| \leq |f(x_2) - f(c)| + |f(c) - f(x_1)| < \varepsilon.$$

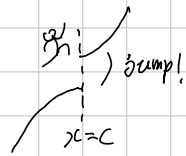
Conversely, take $\delta = \min\{|x_2 - c|, |c - x_1|\}$, then

$$|x - c| < \delta \Rightarrow |f(x) - f(c)| \leq f(x_2) - f(x_1) < \varepsilon.$$

Thus a) $\Leftrightarrow$ b). And, b) $\Leftrightarrow$ c) can be derived by directly:

$\sup\limits_{x < c} f(x) = f(c)$ iff $\forall \varepsilon > 0$, $\exists x < c$ s.t. $f(c) - \varepsilon < f(x) \leq f(c)$

20.5. Let $f:[a,b]\to\mathbb{R}$ be monotonically increasing.



Define a map

$$j_c : (a,b)\to\mathbb{R} ; c \mapsto \inf_{x>c} f(x) - \sup_{x<c} f(x)$$

If $j_c > 0$, we say that f has a jump at the point c.

a) For any $n\in\mathbb{N}$, there exist at most finitely many points at which f has a jump exceeding $\frac{1}{n}$.

More set-theoretically, the following set is finite for all $n\in\mathbb{N}$.

$$D_n = \left\{ c\in(a,b) \,\middle|\, j_c > \frac{1}{n} \right\}$$

b). Deduce that monotone function has at most countively many discontinuous points.

Proof. Let $n\in\mathbb{N}$ be given. Suppose that D_n has infinitely many elements.

Take $m\in\mathbb{N}$ satisfying $n\cdot(f(b)-f(a)) < m$.

Choose $c_1, c_2, \cdots, c_m \in D_n$ satisfying $a = c_0 < c_1 < c_2 < \cdots < c_m < c_{m+1} = b$. Then

$$\sum_{i=1}^{m} j_{c_i} > \frac{m}{n} > f(b)-f(a)$$

Take $\varepsilon > 0$ s.t $\sum_{i=1}^{m} j_{c_i} - \frac{m}{n} > \varepsilon$, and put $r = \frac{1}{2}\cdot\min\{|c_i - c_j| \mid 0\le i,j\le m+1,\ i\ne j\}$.

For each $1\le i\le m$, there exists $r > \delta_i > 0$ s.t

$$\sup_{x<c_i} f(x) - \frac{\varepsilon}{2m} < f(c_i-\delta_i) \le \sup_{x<c_i} f(x) \quad \text{and} \quad \inf_{x>c_i} f(x) \le f(c_i+\delta_i) < \inf_{x>c_i} f(x) + \frac{\varepsilon}{2m}.$$

That is,

$$j_{c_i} = \inf_{x>c_i} f(x) - \sup_{x<c_i} f(x) \le f(c_i+\delta_i) - f(c_i-\delta_i) < \inf_{x>c_i} f(x) - \sup_{x<c_i} f(x) + \frac{\varepsilon}{m} = j_{c_i} + \frac{\varepsilon}{m}$$

Finally, $\sum_{i=1}^{m}\left(f(c_i+\delta_i) - f(c_i-\delta_i)\right) = \sum_{i=1}^{m} j_{c_i} + \varepsilon > \frac{m}{n} > f(b)-f(a)$

But, it is contradiction, because

$$f(b)-f(a) = f(b) - f(c_m+\delta_m) + \sum_{i=1}^{m}\left[f(c_i+\delta_i) - f(c_{i-1}+\delta_{i-1})\right] + f(c_0+\delta_0) - f(a)$$

$$\ge \sum_{i=1}^{m}\left(f(c_i+\delta_i) - f(c_i-\delta_i)\right) \qquad \left(c_i - c_{i-1} \ge 2r > \delta_i + \delta_{i-1}\right)$$

Now, we can prove that b). Since f is continuous at $c\in(a,b)$ iff $j_c = 0$;

$$\{c\in(a,b) \mid j_c \ne 0\} = \bigcup_{n\in\mathbb{N}} D_n \quad \text{is at most countable.} \qquad \blacksquare$$

Note: Since $\mathbb{R} = \bigcup_{n\in\mathbb{Z}}[n,n+1]$, we can extend this statement with respect to $\mathbb{R}$.